Toward Highly Reliable, Precise, and Reproducible Fabrication of Photonic Crystal Slabs on Lithium Niobate

Venancio Calero, Miguel-Angel Suarez, Roland Salut, Bruno Robert, Alexis Caspar, Fadi Baida, Nadège Courjal, and Maria-Pilar Bernal

***Abstract*—Nanostructuring lithium niobate (LN) is recently a hot topic since it can achieve miniaturized footprints in comparison with bulk LN. Due to the challenging machining on this material, the different available techniques still present uncontrolled constraints that forces the LN-based photonic structures behavior to be far from the theory. In this paper, we design a photonic crystal structure based on free standing thin films of LN where the vulnerabilities of design with respect to the fabrication constraints are deeply studied in order to minimize and compensate its impact to the device performances. The developed structures (based on Fano resonances) exhibit a precision in terms of resonance wavelength of $\pm$18 nm and an aspect-ratio decrease of 54.99% accompanied by an outstanding resonant line shape agreement with the theory that is a novel result in nanostructured LN. Furthermore, the implemented resonance takes advantage of the exceptional light confinement provided by the Fano phenomenon enhanced by a band folding effect, exhibiting an outstanding average slope of 0.64 dB/nm. These results demonstrate a step forward in the fabrication of high reproducible, miniaturized, and power efficient devices whose technology can advance the frontiers of the current electrooptical devices.**

***Index Terms*—Electro-optic devices, focused ion beam, lithium niobate, nanostructuring, photonic crystals.**

I. Introduction

PLANAR Photonic Crystals (PhC) have attracted attention for long time because of their potential for miniaturizing device footprints [1], [2]. A specific type of planar-PhC based on Guided Resonances (GRs), commonly implemented through metallic/dielectric gratings [3], [4] or 2D-Photonic Crystal Slabs (PCS) [5]–[7], offers an outstanding coupling with the out-of-plane radiation accompanied by an asymmetric interference pattern due to the Fano phenomenon [5]. The pronounced spectral slope defined by the narrow-band transition between the total transmission and reflection, in addition to the enhanced light-matter interaction [10], reveals unique advantages for the implementation of photonic-based sensors [26], [36], [37]. However, most of works are based on Si or Si_3N_4 which misspend the potential of GRs for implementing E-field or photosensitive devices since these materials do not present notable non-linear (NL) or electro-optic (EO) effects.

For this reason, GRs have been recently studied in lithium niobate (LN) [11], [12] since it presents, in addition to a strong EO response, significant high-order performances. These particularities are also accompanied by wide intrinsic bandwidth, relatively high refractive index ($n_e = 2.13$ at 1550 nm) and large transparency range (from 0.33 μm to 5.5 μm), in contrast with III–V semiconductors where the efficiency is limited by absorption phenomena. The production of LN GR-based PhC requires the use of a $LiNbO_3$ thin film which confines the light vertically in addition with a micrometric size structuring. This micromanufaturing step can be carried out by Proton Exchange (PE)-assisted dry etching [13], laser ablation [14] or assisted chemical etching [15]–[17]. Furthermore, a very promising technique known as Ion Beam Enhanced Etching (IBEE) has demonstrated to provide high resolution and quasi-cilindrical side walls when producing these nanostructures [39]. This technique can be adapted in order to develop PCS based on free-standing LN membranes which provides an additional enhancement of the vertical confinement due to the increased refractive index contrast [18], [19]. Moreover, A.J. Danners group has recently reported quasi-cylindrical holes by Focused Ion Beam (FIB)-milling on LN free-standing membranes produced by IBEE technology [20], [21], pointing out to the fact that this configuration provides a lightening of the sputtered material redeposition. However, the strong light confinement performed in the free-standing GRs also develops highly sensitivity to structural changes [26]. As a consequence, the state-of-art on LN-based PCS generally present a strong deterioration of its resonant features in terms of spectral lineshape and resonance wavelength control as a result from the different fabrication imperfections.

In order to lighten the impact of the imperfections and to achieve high spectral performances, we first employ a non-conventional photonic crystal lattice which exhibits a strong robustness to the conical side walls in addition to offer enhanced

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The authors are with the FEMTO-ST Institute, University of Bourgogne Franche-Comte, Besancon 25000, France (e-mail: venancio.calero@femto-st.fr; miguel.suarez@femto-st.fr; roland.salut@femto-st.fr; bruno.robert@femto-st.fr; alexis.caspar@femto-st.fr; fadi.baida@femto-st.fr; nadege.courjal@femto-st.fr; mpbernal@univ-fcomte.fr).

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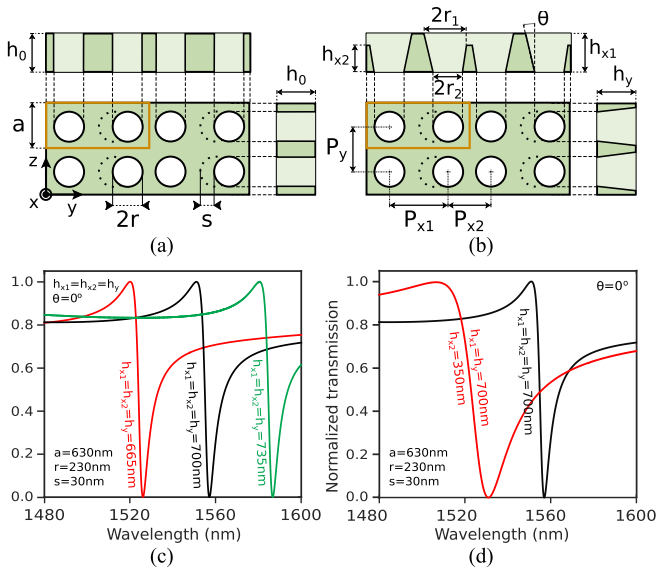


Fig. 1. (a) Sketch of the ideal biperiodical PhC structure. (b) Sketch of a realistic FIB-milled PhC structure regarding its hole conicity and thickness erosion imperfections. (c) Computed transmission spectra comparing the ideal PhC and the thickness-eroded one. (d) Computed transmission spectra considering the difference between h_{x1} and h_{x2}.

light confinement features, as demonstrated in [27]. In addition, we extend this study to the existence of non-uniformities on the slab thickness along the free-standing PCS. Despite this theoretical study can be extrapolated to most of the mentioned nanopatterning techniques, in this work we give special attention to the FIB lithography which is very convenient for prototyping due to its maskless and direct writing features compatible with a broad range of materials. We exploit these advantages by producing the PCS on free-standing ion-sliced membranes enabling highly smooth surfaces [28] which reduces the fabrication constraints to the PhC fabrication itself. Hence, the FIB patterning sustained by a robust numerical analysis of the structure leads to the production of Fano resonances with outstanding spectral performances.

II. Design and Study of the Fabrication Issues

A. Free-Standing Photonic Crystal Slab (PCS)

The sketch of the PCS geometry is represented in Fig. 1(a). The structure consists of an air-filled PhC which is based on a two-dimensional squared lattice, defined by the lattice constant a and the hole radius r. The incident light is set to be normal to the periodicity plane as the PhC provides a phasematching between the external light wavevector and the Guided Modes (GM) of the PCS itself [8], yielding to the GRs. As a particularity of the lattice here employed, the unit cell is enlarged to $2a \times a$ by shifting the even holes a distance s parallel to the crystallographic y-axis. This modification produces a band folding phenomena [29] increasing the light confinement, and avoiding the overlapping of the spectral lineshape with spurious high order modes. The study of the performances and optimization of the structure itself is beyond the scope of this paper, however a detailed study of this structure is reported in [27]. Since the structure is periodical, the slow light is manifested as Slow Bloch Modes (SBM). In this work we focus on the SBMs located at lower frequencies since they lead to reduced structural parameters. As reported in the mentioned work [27], the first two modes, denoted as 1SBM and 2SBM, are also dielectric modes, which are favorable for the enhancement of the sensitivity to changes on the dielectric material [31], [36]. Since most of the available equipment is designed to work at $\lambda_r = 1.55\,\mu$m, the resonant wavelength must fall at its vicinity. The structural parameters fulfilling this requirement are $a = 630$ nm, $r = 230$ nm and $a = 450$ nm, $r = 164$ nm, respectively for the 2SBM and 1SBM.

B. Fabrication Imperfections

In this study we focus on the alterations of the spectral response resulting from the different imperfections coming from the divergence between the ideal and the FIB-patterned structures. Finite Difference Time Domain (FDTD) method is employed considering an infinite PhC structure located in a free space medium. In addition, the source is assumed to be a plane wave and the monitoring conditions correspond to the zeroth diffraction order. More details of the configuration are described in the mentioned work [27]. The most common imperfections produced by the FIB milling can be summarized in:

1) Hole Conicity: The structure presenting the conical holes, frequently referred as sidewall angle and denoted by the angle θ, is sketched in Fig. 1(b). It can be estimated from $\tan\theta = (r_1 - r_2)/h$ where r_1 and r_2 are the hole radius at the top and bottom side of the membrane, while h is the thickness of the slab. It is observed [27] that the structure provides a strong robustness to θ since only a red-shift of the resonance wavelength is produced, nonetheless the deterioration of the aspect-ratio is under the 5% when the angle varies by $\Delta\theta = 2.4°$.

2) Thickness Erosion: Due to the reduced distance between the holes, the gaussian shaped ion beam and the drift of the piezo-electric motor accumulated during the milling, the ions may impact in the holes surroundings, producing inhomogeneities on the slab thickness. In Fig. 1(d), it is observed that the resonant wavelength λ_r is directly proportional to the slab thickness and it has no strong influence on the spectral aspect-ratio. However, if we consider the biperiodical rectangular lattice, the distance between two hole-centers along the y-axis is always $P_y = a - 2r$ as sketched in Fig. 1(b), but in contrast, along the x-axis there are two cases: $P_{x1} = a - 2r + s$ and $P_{x2} = a - 2r - s$. We consider h_0 as the original thickness of the membrane, h_y as the thickness between two holes along the y-direction and h_{x1}, h_{x2}, as the two different thickness along the x-axis, corresponding to P_{x1} and P_{x2}, respectively. Under these conditions, the process may lead to structures for which $h_0 > h_{x1} > h_y > h_{x2}$. The influence of this imperfection to the spectral performances is investigated in Fig. 1(e), where it is observed that the aspect-ratio is notably deteriorated by a broadening of the linewidth, in addition to a strong blue shift of λ_r.

3) Surface Roughness: The sputtering ocurred during the fabrication of the PhC produces the redeposition of the material which leads sometimes to significant surface roughness.

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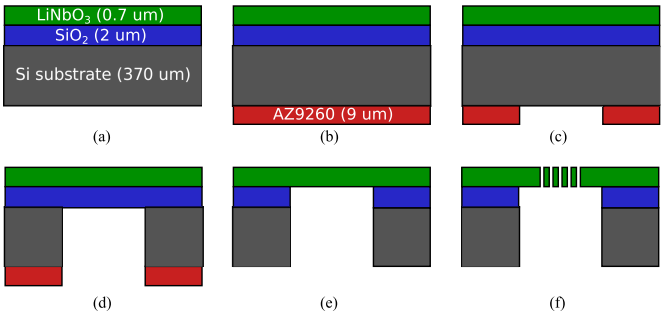


Fig. 2. Fabrication flowchart of the Photonic Crystal Slabs. (a) Composition of the thin layer LNOI stack wafer. (b) Spin coating of the AZ9260 photoresist. (c) Insolation by UV-lithography and resist development. (d) Etching of the silicon substrated by a Bosch process. (e) Wet etching of the BOX layer through HF. (f) Photonic crystal fabrication by FIB milling.

These irregularities may increase of the losses within the slab, broadening the resonance linewidth [32].

III. Fabrication

The implementation of the PCS is formed in two steps: the release of the free standing thin membrane of LN and the fabrication of the PhC within. The fabrication starts from a thin film of LN on insulator (LNOI) stack wafer manufactured by Jinan Jingzheng Electronics Co [33]. The layer distribution from top to bottom consists of a X-cut 700 nm-thick layer of LN, whose thickness has been carefully chosen as a compromise solution between the depth limitation of FIB lithography and the fragility of the layer to intrinsic stresses when being released. This layer relies on a 2 μm-thick SiO_2 buffer oxide layer (BOX). The 380 μm-thick substrate is made of silicon due to its ease to be removed by dry etching processes. The fabrication of the membrane, whose flowchart is depicted in Fig. 2, is performed by standard substrate-based techniques which simplifies the complexity. UV lithography is performed to transfer the 250 μm-diametered circular-shaped membrane to the Si surface. The circular shape of the membranes has been chosen in order to avoid critical stress points that other shapes such as square may produce. For this, AZ9260 photoresist is deposited with a thickness of 9 μm, taking into account the resist selectivity and the 370 μm of silicon to be etched, followed by its insolation and development. The etching of the substrate is performed by a Deep Ion Reactive Etching (DRIE) process which removes the Si volume under the areas where the membrane will be located. The micromasking and non-homogeneities, which may produce silicon grass, are not critical since the BOX is then removed by an Hydrofluoric Acid (HF), providing high tolerance for the DRIE process.

Once the LN thin layer has been released, the free standing membrane must be patterned. In this work, we fabricate both 2SBM ($a = 630$ nm, $r = 230$ nm) and 1SBM ($a = 450$ nm, $r = 165$ nm) structures. A dual beam device, FEI Helios Nano Lab 600i with a Ga+ Liquid Metal Ion Source (LMIS), patterns the PhC within the free standing membrane by employing an acceleration voltage of 30 kV and a probe current of

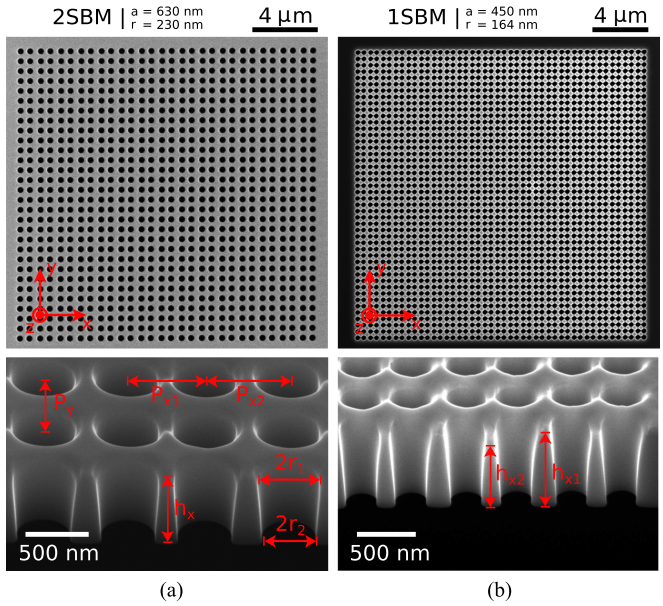


Fig. 3. SEM pictures of the fabricated structure from different views. (a) Top view. (b) Sliced membrane with a SEM tilt of 42°, the section of the PhC is colored in green accompanied by the different structural parameters. (c) Sliced membrane with SEM tilt angle of 42° and horizontally rotated 40° respect to the PhC x-axis.

24 pA. This reduced beam current ensures a negligible surface roughness produced by the sputtering of the surface material. A Elphy Multibeam pattern generator is used to control the beam loops number, step size and dwell time, which are optimized in order to reduce the sidewall angle θ. Furthermore, in order to enhance the patterning precision and to reduce the erosion of the slab thickness, the etching loops are interleaved with drift correction steps based on the correlation between the actual and a reference image of a mark located outside of the PhC area, since the FIB imaging might damage the photonic structure. The implemented PCS, resulting from the described process are reported in Fig. 3(a) and 3(b), for the 2SBM and 1SBM, respectively. It can be observed that the fabricated lattice corresponds properly to the biperiodical photonic crystal structure which leads to a compact area smaller than 19 μm $\times$ 19 μm. A sectional cut of both PCS structures is also shown in Fig. 3. The measured θ values are between 3.6° and 8.5°, however due to the free-standing membrane, the manufactured holes present quasi-cilindrical shape, demonstrating the advantages of nanostructuring on the free standing thin slab in contrast to the tapered holes produced on the bulk [24].

The parameters P_{x1} and P_{x2} can be estimated by computing the autocorrelation of the high resolution SEM images (4096 pixels $\times$ 3775 pixels) in Fig. 3. By the measurement of these values, the asymmetry parameter can be estimated by $s = (P_{x2} - P_{x1})/2$. We have estimated an error on the s parameter up to 7 nm. The thickness erosion phenomenon is noticeable for the 1SBM structure due to its narrow lattice period, which exhibits an structure with $h_0 < h_{x1} < h_{x2} < h_y$. We have estimated h_{x1} and h_y approximately to 650 nm, however h_{x2} is strongly thinned downto values between 600 nm and

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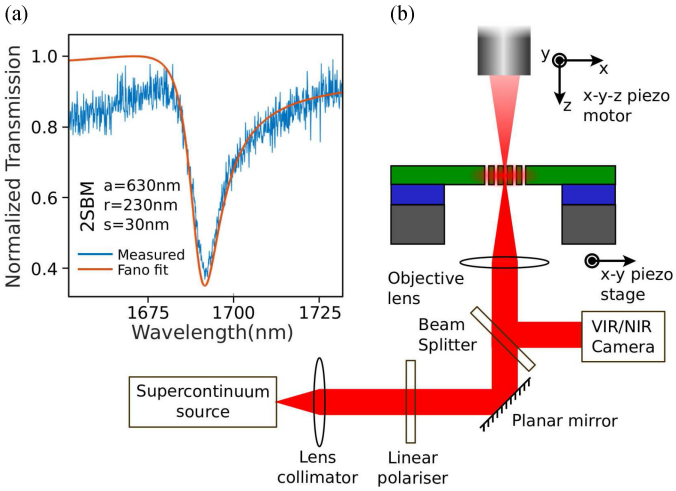


Fig. 4. (a) Measured transmission spectrum of the PCS with $a = 630$ nm, $r = 230$ nm, $s = 30$ nm and $N = 30$ in blue. A fit of the Fano formula is represented in orange. (b) Sketch of the experimental system employed for the transmission characterization of the PCS.

400 nm as observed in Fig. 3(b). Since the inhomogeneities on the slab thickness may have a strong impact on the resonance aspect-ratio (see Fig. 1(a)), in this work we focus on the 2SBM structure that leads to $h_0 \approx h_{x1} \approx h_{x2} \approx h_y$, reducing the fabrication imperfections to the sidewall angle θ.

IV. Optical Characterization

The fabricated structure presents ultra compact dimensions ($19\,\mu$m $\times$ $19\,\mu$m $\times$ $0.7\,\mu$m) which turns to compulsory high precision light injection and collection systems. In addition to this, the GRs here employed are polarization sensitive which means that the injection must also provide an accurate control of the light polarization. The characterization system is depicted in Fig. 4. It is fed by a supercontinuum laser source (LEUKOS-SM-OEM) with an spectral range of 320-2400 nm in order to provide a broad spectral characterization. The pump source of the laser emits at 1064 nm, with a pulse width shorter than 1 ns, a repetition rate of 20 kHz and a mean power of 60 mW. This pulsed source leads to a high peak power, in the order of kW, that may produce the ablation of the PhC structure when focusing on in the thin membrane. For this reason, prior to the inverted microscope, light is attenuated to ensure the peak power to be under the damage threshold. Then, light is linearly polarized and a half-wave plate can be also added for higher accuracy at the alignment with the y-axis of the PhC. The light injection is performed by a microscope objective (10x, NA = 0.25), whose characteristics have been chosen to provide a spot size large enough to excite the GRs. In order to increase the spatial accuracy of the injection into the PhC, the sample holder position is controlled by a dual-axis x-y piezo stage with nanometric resolution, accompanied by a VIR/NIR camera to visualize the location of the laser spot with respect to the center of the PhC. In order to collect the light, a single mode fiber (SMF-28e) is positioned next to the PCS. Diffraction effects may occur due to the finite dimensions of the structure which is designed to optimize the Fano spectral aspect-ratio for the zeroth order [27]. In order to avoid collecting spurious diffraction orders, the fiber is positioned by a three-axed piezoelectric motor with nanometric precision. In addition to this, the Fabry Perot effect produced between the PCS and the fiber facet may be strong enough to hide the Fano resonance. For this reason, the z-axis of the piezo is relevant, since the nanometric control allows a decrease of the distance to few micrometers, increasing the free spectral range (FSR).

TABLE I
Summary of the GRs Performances

Ref	Configuration	Mean slope	Slope fall	λ_r error
[11]	Dielectric grating	0.063 dB/nm	92.54 %	< 100 nm
[12]	Metallic grating	0.027 dB/nm	56.60 %	< 60 nm
[20]	Free-standing PCS	0.078 dB/nm	65.31 %	< 50 nm
[21]	Free-standing PCS	0.096 dB/nm	82.98 %	< 50 nm
This work	Free-standing PCS	0.643 dB/nm	54.99 %	< 18 nm*

*Error after applying the compensation described in Section V.

The light, collected by the fiber, is introduced into an optical spectrum analyzer (OSA) which measures the transmission spectrum depicted in Fig. 4(a). In order to eliminate the different spectral features introduced by the elements within the experimental system, the measured spectrum has been normalized with respect to the cross polarization since the resonance is located in a different wavelength due to the ordinary index ($n_o = 2.211$ at 1550 nm). A fit of the Fano formula [34] to the experimental results demonstrates a good agreement with the theoretical resonance lineshape. The average slope of the resonance lineshape is 0.64 dB/nm, demonstrating a deterioration of 54.99% respect to the ideal computed structure represented in Fig. 1(a). In comparison with the GRs-based structures summarized in Table I, our PCS presents an outstanding spectral slope in addition to the lowest deterioration with respect to the ideal numerical structure. Despite the low deterioration in terms of spectral lineshape, the control of the resonant wavelength λ_r is poor since this result presents $\lambda_r = 1690$ nm, exhibiting an error of $\varepsilon(\lambda_r) = +131$ nm. This inaccuracy on the resonant wavelength can be overcome by taking care of the remaining imperfection: the sidewall angle.

V. Compensating the Imperfections

Despite of the fact that the thickness erosion is negligible for the fabricated structure and the hole conicity is reduced with respect to bulk crystal fabrication, the sidewall angle still presents a value higher than 0°. As this geometrical imperfection produces a red shift on λ_r of 13.3 nm$/^\circ$ [27], taking into account the measured sidewall angles θ, the introduced deviation can be estimated up to $+113.1$ nm. This main contribution to $\varepsilon(\lambda_r)$ can be compensated by rescaling the structural parameters down [35].

A first result is shown in 5(a) where we decrease the structural parameters to $a = 600$ nm and $r = 219$ nm, leading to a blue shift which centers the resonance wavelength at $\lambda_r = 1606$ nm and verifying this method for the compensation of the conicity. In order to set the resonance nearby $\lambda_r = 1550$ nm, the structural parameters scaled down to $a = 578$ nm and $r = 211$ nm. This scaled geometry leads to the transmission spectrum represented in Fig. 5(b), where it can be determined $\lambda_r = 1553$ nm

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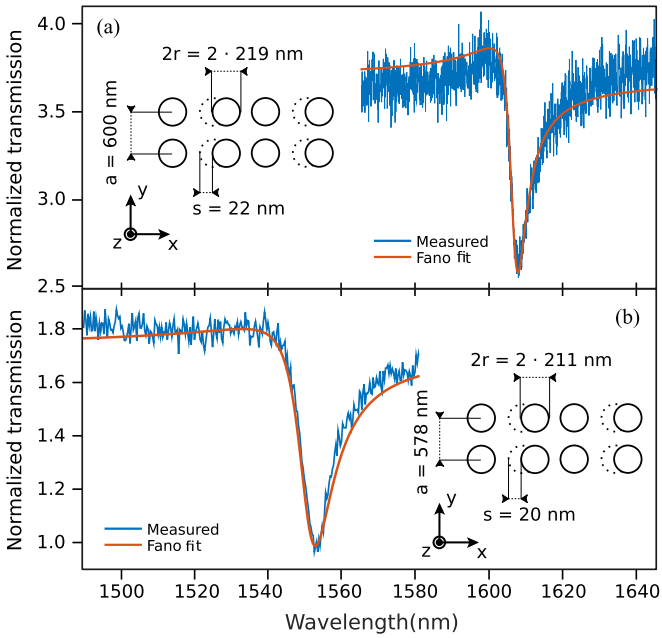


Fig. 5. Measured spectra from the compensated geometry PhC. (a) $a = 600$ nm, $r = 219$ nm, $s = 19$ nm and $N = 30$. (b) $a = 578$ nm, $r = 211$ nm, $s = 30$ nm and $N = 30$.

and the left slope, the most pronounced, successfully contains the wavelength $\lambda = 1550$ nm being suitable for setting there the working point λ_w of the final device. Furthermore, by observing a set of tens of samples whose PhC is fabricated by keeping fixed the re-scaled parameters for FIB milling, we observe a maximum deviation around the target working wavelength of ± 18 nm. This value can be attributed mainly to the error on the thickness of the LN layer, estimated by the manufacturer to ± 10.4 nm, that according to the study in Fig. 1(d), it introduces an error of ± 8.9 nm on λ_r. In addition to this, the human-dependent FIB features such as the beam focus and the aperture which may have slightly altered between two different fabrication processes, adding a second source of error to the resonant wavelength.

VI. Conclusion

We have optimized the procedure for the fabrication of PCS based on free-standing membranes of LN being aware of the different issues which are commonly introduced during the PhC fabrication, in our case by FIB lithography. The surface roughness is negligible since the probe current is set to low enough (24 pA). Also, the designed structure presents a lattice constant large enough to get rid off the thickness erosion, leading to a simpler problem where the hole conicity is the main factor altering the nanostructure optical behavior. The holes exhibit a quasi-cylindrical shape demonstrating the advantageous configuration of the free standing PCS. The control on the resonant wavelength λ_r has been achieved, leading to a reproducibility error lower than ± 18 nm. In addition, this method leads to a spectral aspect-ratio deterioration of 54.99%, which is lower in comparison with previous works and also, to a high slope of 0.643 dB/nm which is also the highest reported through guided resonances on LN, being interesting in terms of low power consuming and high sensitive features. This progress on the control of the reliability of the fabrication is very powerful for developing high accurate and compact optical components such as sensors (temperature, E-field or refractive index sensing), electro-optical modulators or other PhC-based devices.

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